



Ultrasound guidance for neuraxial anesthesia techniques

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Literature review current through: **Jul 2026**.

This topic last updated: **Oct 09, 2024**.

INTRODUCTION

Neuraxial anesthesia techniques (spinal, epidural, and combined spinal epidural) are typically performed by using surface anatomic landmarks to guide needle placement, which is occasionally challenging. Neuraxial ultrasound has emerged as a noninvasive and safe tool that may predict difficulty with the neuraxial procedure, and may improve accuracy and success rate, particularly in patients with difficult anatomy.

This topic will discuss relevant spine anatomy and ultrasound correlates, provide a step-by-step approach to ultrasound scanning, and appraise the evidence for benefits of neuraxial ultrasound.

Epidural and spinal anesthesia techniques and the basics of ultrasound for regional anesthesia are discussed separately.

- (See ["Spinal anesthesia: Technique"](#).)
- (See ["Epidural and combined spinal-epidural anesthesia: Techniques"](#).)
- (See ["Ultrasound for peripheral nerve blocks"](#).)

UTILITY OF NEURAXIAL ULTRASOUND

Neuraxial anesthesia techniques are usually performed by palpating bony landmarks. As an alternative or in addition, neuraxial ultrasound may be used to determine the intervertebral level for neuraxial needle placement, identify the midline and optimal angle of needle insertion, and measure the depth to the epidural and intrathecal space. Ultrasound can also help assess anatomy that may make neuraxial procedures challenging (eg, spine rotation), identify the most accessible interspace, and confirm existing spinal hardware. This is especially useful if radiologic spinal imaging is not available.

- The intervertebral level (eg, L2-L3, L3-L4, L4-L5, etc) can be identified using the parasagittal oblique interlaminar view. Options include visualizing the sacrum in the lumbar spine or the articulation of the transverse process with the 12th rib in the thoracic spine, counting vertebrae while moving the transducer cephalad in a parasagittal orientation. For high thoracic procedures, ribs can be counted down from the first rib (T1). (See ["Parasagittal oblique interlaminar view"](#) below.)
- The midline can be identified by scanning in transverse orientation and visualizing the spinous process or interspinous window which includes the posterior complex (ligamentum flavum/dura). (See ["Transverse scan"](#) below.)
- Depth to the ligamentum flavum/dura complex can be measured with the caliper function on the ultrasound screen with the transducer in either the parasagittal oblique or transverse interspinous orientation. (See ["Parasagittal oblique interlaminar view"](#) below and ["Interspinous view"](#) below.)
- Spine rotation can be assessed by visualizing the symmetry of the height of the articular processes in the transverse interspinous view. (See ["Interspinous view"](#) below.)

- Optimal needle insertion position can be determined by locating the intervertebral level that contains the best view of the posterior and anterior complexes in the transverse interspinous view. Once this is found, the angle required to achieve this view is mimicked with actual needle insertion. (See ['Interspinous view'](#) below.)

Mid to upper thoracic neuraxial ultrasound is more challenging than lumbar and low thoracic neuraxial ultrasound, however several useful pieces of information can be obtained. The parasagittal oblique interlaminar view can be used to identify the correct intervertebral level and to estimate a safe needle insertion depth by measuring the distance to the laminae and interlaminar space. Despite the narrower interlaminar window, the posterior and/or anterior complex can frequently be visualized with the transducer in this orientation. However, transverse interspinous views are usually not possible in the mid to upper thoracic spine because the spaces between spinous processes are very narrow. Unlike lumbar scans, the posterior and anterior complexes are infrequently seen as hyperechoic bands within the space.

Evidence regarding benefits and outcomes of neuraxial ultrasound is discussed below. (See ['ultrasound guidance versus landmark based neuraxial techniques'](#) below.)

ANATOMY

- **Bones** – There are five lumbar and 12 thoracic vertebrae, each consisting of a vertebral body and arch ([figure 1](#)). The arch consists of a pair of pedicles and laminae from which the spinous process, transverse processes, and superior and inferior articular processes arise.
 - The spinous processes, when viewed laterally, project posteriorly and inferiorly and are broad and rectangular in shape. In the lower thoracic spine (T10 to T12), the spinous processes and other vertebral anatomy are similar to the lumbar spine. In the upper to mid thoracic spine (T4 to T9) the spinous processes are steeply angled more caudally, and the spaces between them (ie, the acoustic window for ultrasound imaging) are narrower ([figure 2](#)).
 - The transverse processes are thin, wing-like, bony extensions from the superior part of the vertebral body that project laterally and incline slightly posteriorly.
 - Laminae are broad, bony surfaces connecting the transverse processes to the spinous processes. In the lumbar area, they slope anterior to posterior, following a cephalad to caudad direction. Laminae in the thoracic spine are flatter and overlapping, producing smaller interlaminar windows to visualize neuraxial structures.
 - The superior and inferior articular processes are small, bony structures that articulate with adjoining vertebrae; the articulation between each pair is called a facet or zygapophyseal joint. In the upper-to-mid thoracic spine, the articular processes are less prominent posteriorly and are therefore not easily seen with neuraxial ultrasound.

The spinal cord is situated within the vertebral canal, which is bordered ventrally by the posterior aspect of the vertebral bodies and dorsally by lamina and spinous process.

- **Ligaments** – The spinal column is stabilized by multiple ligaments ([figure 3](#)).
 - The supraspinous ligament runs continuously along the posterior midline aspect of the spine and connects the apices of the spinous processes.
 - The interspinous ligament connects adjoining spinous processes and blends anteriorly and posteriorly with the ligamentum flavum and supraspinous ligament, respectively.
 - The ligamentum flavum connects the laminae of adjacent vertebrae and is an important landmark for neuraxial techniques.
 - The anterior and posterior longitudinal ligaments connect the anterior and posterior surfaces of vertebral bodies.
- **Thecal sac** – The thecal/dural sac rests within the canal and contains the spinal cord and cerebrospinal fluid ([figure 4](#)). The dura mater is a dense connective tissue external to the arachnoid and pia that encapsulates and protects the spinal cord and brain.
- **Epidural space** – The epidural space consists of three components: the posterior, anterior, and lateral epidural compartments ([figure 4](#)).

- The posterior compartment is frequently accessed for neuraxial anesthetics and chronic pain procedures (eg, epidural steroid injections). It is bordered anteriorly by posterior dura and posteriorly by the ligamentum flavum.
- The lateral epidural space extends to the pedicles and intervertebral foramina.
- The anterior epidural space contains the internal vertebral venous plexus and posterior longitudinal ligament and is bordered anteriorly by the anterior dura.

The epidural space is generally largest at the midline and narrows laterally. It is also largest in the mid-lumbar spine and becomes narrower in both the thoracic spine and sacral regions.

ULTRASOUND IMAGING

Patient positioning — Neuraxial ultrasound is performed after positioning the patient for the neuraxial anesthesia procedure. Patients can be positioned in the lateral or sitting position for lumbar neuraxial procedures, and are almost always positioned sitting for thoracic epidural placement if the patient is able. (See "[Epidural and combined spinal-epidural anesthesia: Techniques](#)", section on 'Positioning for epidural procedure'.)

For neuraxial ultrasound, we suggest using the sitting position for almost all procedures, particularly when learning neuraxial ultrasound. The sitting position may facilitate understanding of spatial orientation and improve ergonomics.

Maximizing flexion of the lumbosacral spine can greatly improve acoustic windows and the path for needle insertion ([figure 5](#) and [figure 6](#)). One method for achieving this in the sitting position is to tilt the procedural table laterally to elevate the patient's knees relative to the buttocks. This causes the patient to naturally lean forward and flex the spine ([picture 1](#)). Commercial devices are available that are designed to maximize flexion of the spine while seated during neuraxial block ([figure 7](#)).

Equipment — A low frequency (2 to 5 MHz), curvilinear transducer is recommended for neuraxial ultrasound. Curvilinear transducers obtain a broad field of view. Low frequency transducers are required for visualizing deeper structures, though they provide lower resolution images. Preprocedure ultrasound is usually performed prior to skin preparation. However, if difficulty is encountered during the scan (moderate degrees of spinal rotation or extremely deep structures), sterile ultrasound gel and a sterile ultrasound transducer cover sheath can be used to allow minor adjustments of the initial (scout) scan during the neuraxial procedure.

The depth should be set at 7 to 10 cm. By convention and to assist with image recognition, the transducer is oriented so that the left side of the ultrasound screen is cephalad.

Sonoanatomy and transducer orientation — Bony structures such as spinous processes, articular processes, and transverse processes do not transmit ultrasound waves well and thus appear as hyperechoic lines (white) with black acoustic shadowing underneath. Ligaments such as the ligamentum flavum and posterior longitudinal ligament appear hyperechoic (white), cerebrospinal fluid appears anechoic (black), and muscles generally appear hypoechoic (dark gray).

The three transducer orientations used for neuraxial ultrasound are parasagittal, parasagittal oblique, and transverse ([figure 8](#)). A diagonal orientation (combined parasagittal and transverse) has been described for real-time neuraxial ultrasound [1]. (See '[Real-time neuraxial ultrasound](#)' below.)

The images used for neuraxial ultrasound are discussed here. When learning to perform neuraxial ultrasound, we suggest obtaining these views in sequence, as described here and in the step-by-step sequence discussed below. Once the clinician is comfortable with neuraxial ultrasound, it is reasonable to start immediately with the parasagittal oblique interlaminar view. (See '[Step-by-step lumbar ultrasound](#)' below and '[Step by step thoracic ultrasound](#)' below.)

Parasagittal scan — The parasagittal scans are used to identify the location of the laminae (in medial-lateral orientation) and ultimately the interlaminar space. Recognition of the laminae is often easiest when its morphology is compared with that of more lateral vertebral structures (articular processes and transverse processes). Pattern changes are subtle and only within a few centimeters of each other therefore slow deliberate scanning is recommended.

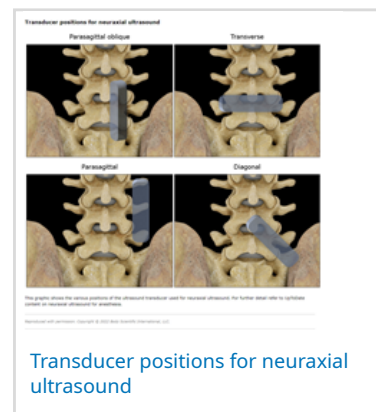


Figure 8 - larger image below

Transverse process view — For this view, place the transducer 4 to 5 cm lateral to the spinous process in the parasagittal plane ([figure 9](#)). Transverse processes are seen as hyperechoic, crescent-shaped reflections with hypoechoic, finger-like acoustic shadows below. This creates a pattern that has been termed the "trident," sign because of its similarity to the mythologic weapon. The psoas muscle has a striated appearance and is located in between the transverse processes; the erector spinae muscle lies superficial to the transverse processes.

Articular process view — For this view, obtain the transverse process view, then slide the transducer medially while maintaining the parasagittal transducer orientation ([figure 10](#)). As the transducer is moved, the finger-like acoustic shadows of the transverse processes transition to a continuous, scalloped hyperechoic line formed by the overlap of superior and inferior articular processes, at approximately 2 to 3 cm lateral to the spinous processes. The pattern visualized is termed the "camel hump," sign.

Parasagittal oblique interlaminar view — The parasagittal oblique interlaminar scan is used to determine the intervertebral level and measure the depth to the interspace. The interlaminar space is easily visualized in this view and contains the posterior complex (ligamentum flavum/dura) and anterior complex (posterior longitudinal ligament/vertebral body). Posterior and anterior complexes cannot be visualized in the articular process or transverse process views ([figure 11](#)).

For this view, obtain the articular process view, then slide the transducer slightly medially, and angle the ultrasound beam 5 to 10 degrees towards the midline. As the ultrasound beam is moved towards the neuraxis, the "camel hump," pattern will morph into a succession of sharp "sawtooth"-shaped structures, which represent lamina. The sacrum can be seen as a horizontal, slightly wavy hyperechoic line with a dorsal tilt and dense acoustic shadow underneath. The soft gaps in between the triangular "sawteeth" represent interlaminar spaces.

Within the gap, two horizontal hyperechoic bands resembling an "equal sign," can be seen with a large black space (the thecal sac) in between. These two bands are the posterior and anterior complexes. Though spinal segments such as the conus medullaris and cauda equina are easily visualized in infants due to normal incompletely ossified posterior structures at this age, these components are not seen in adults.

- **Posterior complex** – The shallowest posterior hyperechoic band, visualized near the lower edge of the sawtooth (lamina), is called the posterior complex. The posterior complex consists of the ligamentum flavum, epidural space, and posterior dura and may be visualized collectively as a single structure. Sometimes, especially in patients with lower body mass index, the ligamentum flavum and posterior dura may be seen as two thin closely approximated hyperechoic parallel lines separated by the hypoechoic epidural space.
- **Anterior complex** – The anterior complex is a second, thicker horizontal hyperechoic band, approximately 1 to 1.5 cm anterior to the posterior complex, and is the deepest visualized structure. The anterior complex consists of the anterior dura, anterior epidural space, posterior longitudinal ligament, and posterior aspect of the vertebral body. Like the posterior complex, these structures are often closely approximated and are most often seen as a single hyperechoic linear structure. The anterior complex is usually thicker, brighter, and more easily seen than the posterior complex.

In the mid and upper thoracic spine, the laminae appear as overlapping flat structures, similar in appearance to roof shingles. The interlaminar spaces are seen as narrow hypoechoic interruptions. Unlike at the lumbar levels, the posterior and anterior complexes are only occasionally seen as hyperechoic bands within the space ([figure 12](#)).

Transverse scan — The transverse scans can be used to identify midline, measure the depth to a target interspace, and assess for rotation of the spine (eg, in scoliosis).

Spinous process view — Place the transducer in a transverse orientation over the spinous process at the desired interspace ([figure 13](#)). The tip of the spinous process appears as a superficial hyperechoic semi-circle with large acoustic dropout below. The erector spinae muscles are visualized on either side of the spinous process. Just adjacent to midline, the laminae appear as two thick horizontal hyperechoic lines and cast a large hypoechoic shadow anteriorly (deeply).

In the mid and upper thoracic spine, the laminae are seen as horizontal hyperechoic structures deeper and just adjacent to the spinous process ([figure 14](#)). The transverse processes articulate with the rib tubercles and appear laterally as wing-like, hyperechoic structures.

Interspinal view

- **Lumbar** – Obtain this view by sliding the transducer caudal or cephalad from the spinous process view, between spinous processes ([figure 15](#)). The interspinous ligament appears as a hypoechoic vertical column in the midline. In this view, the pattern of shadows created by the bones resembles a bat (mammal) and is referred to as the "bat sign" ([image 1](#)).
- The articular processes form the ears and are the most superficial structures of the bat. They appear as hyperechoic, symmetrical, cap-like structures with black acoustic shadowing below. Articular processes present at uneven heights may indicate spine rotation, which can be due to patient positioning or scoliosis (or may represent transducer rotation).
- The transverse processes lie far lateral and deep to the articular processes. They appear hyperechoic and flat, with an upwards posterior tilt resembling the wings of the bat.
- The hyperechoic line connecting the articular processes (top of the bat's head) is the posterior complex. Like its appearance in the parasagittal oblique view, this horizontal band can appear as one or two closely approximated horizontal lines representing the ligamentum flavum, epidural space, and posterior dura.
- The intrathecal space is seen as an anechoic space directly below the posterior complex.
- The anterior complex (bat's bottom) is a thick hyperechoic horizontal deep line, and due to its brightness, is often an easily recognizable structure.

Note: Importantly, the anterior complex always lies deep (anterior) to the transverse processes, whereas the posterior complex usually lies in the same horizontal plane as the midpoint of the transverse process. These relationships are important because the anterior complex may be mistaken for the posterior complex in difficult scans. This can lead to overestimation of the depth to the dura, and can potentially result in an inadvertent dural puncture.

- **Thoracic** – The transverse interspinous view is usually challenging to obtain in the thoracic level and often requires more cephalad angulation of the transducer. However, when obtained, the anterior complex is easier to visualize than the posterior complex, and the transverse processes are easily visualized. The articular processes are less prominent posteriorly in the thoracic level, so the characteristic "bat shape" is not observed.

STEP-BY-STEP LUMBAR ULTRASOUND

We suggest using this sequence when learning to perform neuraxial ultrasound. Clinicians who are comfortable with neuraxial sonoanatomy may want to start by obtaining a parasagittal oblique interlaminar view (step 4).

- **Step 1: Position the patient** – Position the patient for the neuraxial procedure, and optimize flexion of the spine. Palpate the spinous processes. (See '[Patient positioning](#)' above.)

For best ergonomics, scan with the dominant hand. In patients with higher body mass index, ideally brace the forearm against the bed or patient's back to maintain pressure on the transducer.

Tip: After positioning, ask the patient to remain still throughout the ultrasound and neuraxial anesthesia procedure. If the patient must move after scanning and skin marking, it may be prudent to scan again using sterile gel and a transducer sheath prior to needle insertion, adjusting skin markings as necessary.

- **Step 2: Obtain a parasagittal transverse process view** – Place a low frequency curvilinear transducer longitudinally in a parasagittal orientation 4 to 5 cm lateral to the midline to obtain the parasagittal transverse process view ([figure 9](#)). (See '[Equipment](#)' above and '[Transverse process view](#)' above.)

Transverse processes are seen in a "trident sign" pattern.

- **Step 3: Obtain a parasagittal articular process view** – Keeping the transducer in the same orientation, slide the transducer medially (1 to 2 cm lateral to midline) to obtain the parasagittal articular process view ([figure 10](#)). (See '[Articular process view](#)' above.)

Articular processes are seen in a "camel hump," pattern with no intervening gaps.

- **Step 4: Obtain a parasagittal oblique interlaminar view** – Slide the transducer slightly medially and angle the beam 5 to 10 degrees towards the midline to obtain the parasagittal oblique interlaminar view ([figure 11](#)).

Laminae are seen in a "sawtooth," pattern. The anterior and posterior complexes can be visualized in the soft gaps representing the interlaminar spaces.

- **Step 5: Identify desired interspace** – Slide the transducer caudally if necessary to visualize the sacrum, which will appear as a horizontal, often slightly wavy hyperechoic line. The first sawtooth represents the lamina of L5 ([figure 11](#)). The hypoechoic gap between the sacrum and lamina represents the L5-S1 interspace.

Center each hypoechoic interspace in the midline on the ultrasound screen. On many ultrasound machines, this process can be facilitated by turning on the "M-mode," "Centerline" or "M/D cursor," function which produces a vertical centered line on the ultrasound screen ([image 2](#)). Make a horizontal mark on the skin adjacent to the midpoint of the long edge of the transducer. Continue sliding the transducer cephalad, marking each interspace until the desired interspace is reached ([figure 16](#)).

- **Step 6: Measure the depth to the posterior complex** – Use the caliper setting to measure depth to the posterior complex ([image 3](#)). For accurate estimations of depth, gently release pressure off the transducer before measuring.

- **Step 7: Obtain a transverse interspinous view** – Rotate the transducer 90 degrees horizontally and medially to obtain the transverse interspinous view with the "bat sign," pattern ([figure 15](#) and [image 1](#)). If a transverse process spinous process view is obtained ([figure 13](#)), move the transducer slightly cephalad or caudad in the direction of the skin mark towards the desired interspace. Angling the ultrasound beam 5 to 10 degrees cephalad may improve visualization of the "bat sign."

Closely examine the "bat sign," and identify the posterior complex (top of bat's head) and anterior complex (bat's bottom). The transverse processes (wings of the bat) are the hyperechoic structures most lateral on ultrasound. Ensure that the posterior complex is correctly identified by ensuring that the anterior complex is deep (more anterior) to the transverse processes. Occasionally, there is poor visualization of the bat at the desired interspace. Slide the transducer one level cephalad or caudad and select the level with the clearest view of the posterior and anterior complexes and that which contains articular processes (bat's ears) at the most even heights.

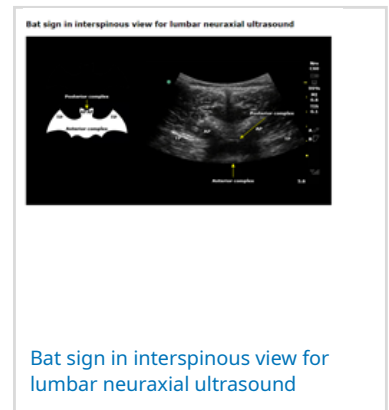


Image 1 - larger image below

- **Step 8: Identify the midline** – Center the image on the ultrasound screen, using the "M-mode," "Centerline" or "M/D cursor," function (if available) to assist, and making slow adjustments rocking the transducer clockwise and counterclockwise and side to side as necessary. Once the articular process heights are equal and the image is centered, make a vertical skin mark adjacent to the midpoint of the long side of the transducer. Confirm that the horizontal mark made previously in the parasagittal oblique view is centered with the short end of the transducer. Make small adjustments of the horizontal line if necessary. The intersection of the horizontal and vertical lines is the target for needle insertion ([figure 17](#)).
- **Step 9: Measure the depth to the posterior complex** – Measure needle insertion depth (skin to posterior complex) by gently releasing pressure off the transducer and using the caliper function ([image 4](#)). Depth in this view should be similar to that obtained in the parasagittal oblique view.
- **Step 10: Note the angle for needle insertion** – Prior to removing the transducer, memorize the angle of the ultrasound transducer that produced the clearest interlaminar view. The needle insertion should closely mimic any cephalad or left-to-right tilt of the transducer.

STEP BY STEP THORACIC ULTRASOUND

The following modified paramedian approach applies to mid to upper thoracic (T4 to T9) neuraxial ultrasound. Lower thoracic ultrasound is similar to lumbar ultrasound in that a midline approach can usually be used successfully.

- **Step 1: Position the patient** – This step is the same as it is for lumbar ultrasound, above.
- **Step 2: Identify the correct interspace** – Obtain a parasagittal view 5 cm lateral to the midline, to identify the 12th rib. Visualize the intercostal muscles and the shimmering appearance of the pleura between ribs. Slide the transducer cephalad, counting from the 12th

rib to find the desired interspace ([figure 18](#)). Alternatively, the desired interspace can be identified by counting up in a parasagittal oblique view from the sacrum (similar to lumbar neuraxial ultrasound).

For high thoracic procedures the ribs can be counted down from the first rib. Locate the first rib can by placing the transducer in a parasagittal orientation in the supraclavicular fossa. Maintain a parasagittal orientation and slide the transducer over the shoulder near the base of the neck, staying medial to the scapula. Visualize the second rib on the patient's upper back. Slide the transducer caudally, counting ribs downwards to the desired level.

- **Step 3: Obtain a parasagittal transverse process view** – Slide the transducer 1 to 2 cm medially from the rib view to obtain a parasagittal transverse process view ([figure 19](#)). The transition of rib to transverse process is subtle and occasionally difficult to appreciate. Ribs have a rounder contour with easily identifiable sliding pleura in between; transverse processes are more rectangular and intervening pleura is more difficult to visualize.
- **Step 4: Obtain a parasagittal oblique interlaminar view** – Once the transverse process view is obtained, move further medially and obliquely to obtain a parasagittal oblique interlaminar view ([figure 12](#)). The appearance of the laminae is distinctly different from the lumbar region in that they are flat and overlapping (similar to roof shingles) with very small intervening gaps. Choose the level with the widest interlaminar space with the clearest view of the posterior and anterior complexes. If a poor view is obtained or if scoliosis is present, it can be useful to scan the other side of the spine. Center the narrow interlaminar space in the midline on the ultrasound screen. This process can be facilitated by turning on the "M-mode," "Centerline," or "M/D cursor," function which produces a vertical centered line on the ultrasound screen. Make a horizontal mark on the skin adjacent to the midpoint of the long edge of the transducer.
- **Step 5: Measure the depth to the laminae** – If the posterior complex is visualized, gently release pressure off the transducer and use the caliper function to measure the depth from the skin to the posterior complex. Given that this measured depth is perpendicular from the skin, it is an underestimation of the actual needle depth from the skin to the space, often by several centimeters. This is because the epidural needle is angulated cranially in the thoracic region. If the posterior complex is not visible, measure the depth from the skin to the laminae. This depth can be an extremely useful guide because it represents the minimum safe depth required to hit the laminae with the needle perpendicular to the skin ([figure 12](#)).
- **Step 6: Identify the midline** – Rotate the transducer 90 degrees to horizontally and medially to obtain a transverse interspinous or transverse spinous process view, at the level of the horizontal mark for the desired interspace ([figure 14](#)). Center the image on the ultrasound screen, using the M-mode function (if available) to assist. Make a vertical skin mark adjacent to the midpoint of the long side of the transducer.
- **Step 7: Insert the epidural needle** – The intersection of the horizontal and vertical lines represents the target for the needle tip entry into the epidural space. However, because of the narrow interlaminar space and overlapping laminae, a paramedian needle approach is best in the upper to mid thoracic spine, with needle insertion ([figure 20](#)) approximately 0.5 cm lateral to and 1.5 cm inferior to the intersection of the horizontal line and vertical lines marked in previous steps. Insert the needle perpendicularly at this point to hit lamina, taking into account the safe distance calculated above. Slowly walk off the lamina angling caudally until the interlaminar space is reached. Unlike traditional landmark-based paramedian approaches which use larger angulations medially (15 to 20 degrees), medial angulation of the needle to enter the space is minimal with this approach (5 degrees is generally sufficient).

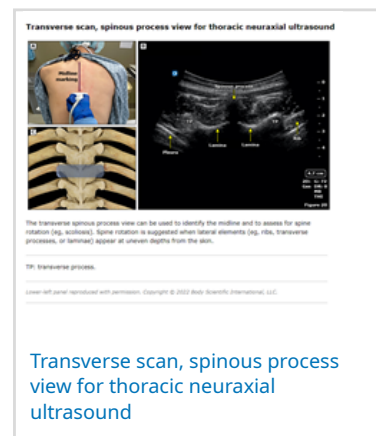


Figure 14 - larger image below

ULTRASOUND GUIDANCE VERSUS LANDMARK BASED NEURAXIAL TECHNIQUES

Compared with standard landmark-based lumbar neuraxial procedures, preprocedure ultrasound can increase the success and reduce the technical difficulty of neuraxial procedures [2]. Ultrasound can more accurately determine the spinal level for block. It can be used to estimate the depth from the skin to the dura, confirm the midline, and identify the most accessible interspace and the optimal angle for needle insertion, and can be used to assess for spine rotation. (See 'Utility of neuraxial ultrasound' above.)

Theoretical but unproven benefits of neuraxial ultrasound include preventing spinal cord injury by identifying the correct vertebral level, and reducing the risk of spinal epidural hematoma, by reducing the number of attempts and bloody taps. The existing studies are not large enough to determine whether neuraxial ultrasound confers a safety benefit, because of the very low incidence of major complications (eg, spinal epidural hematoma, traumatic spinal cord injury) with neuraxial anesthesia.

However, neuraxial ultrasound takes time to perform and requires training to achieve proficiency [3]. We suggest that clinicians use preprocedure ultrasound on a regular basis, to develop competency with normal anatomy before relying on its use during difficult procedures.

Most studies comparing preprocedure ultrasound with landmark based techniques have involved lumbar neuraxial procedures.

Neuraxial procedure efficacy — Preprocedure ultrasound may decrease the likelihood of failed neuraxial procedures (inability to place, need for replacement due to incomplete analgesia, multiple pass attempts). Examples of relevant studies include the following:

- In a 2013 meta-analysis of 12 randomized trials (1234 patients) who underwent spinal or epidural anesthesia procedures or diagnostic lumbar puncture, neuraxial ultrasound reduced the incidence of a failed neuraxial procedure by 79 percent (1 versus 7 percent, risk ratio 0.21, 95% CI 0.10-0.43) [4]. This meta-analysis included three trials using real time ultrasound and one trial in pediatric patients.
- In a subsequent meta-analysis of 13 trials (1678 patients) of adults who underwent neuraxial procedures, preprocedure ultrasound reduced the incidence of technical failure by half (2.8 versus 5.8 percent, risk ratio 0.51, 95% CI 0.32-0.8) [2]. This meta-analysis included six of the same trials that were in the meta-analysis described above [4].
- One of the trials included in both meta-analyses described above included 370 parturients who had labor epidurals placed by first year anesthesia residents [5]. The use of preprocedure ultrasound resulted in lower incidence of failed epidural both within 90 minutes of placement (1 versus 6; 0.5 versus 3.3 percent), and also more than 90 minutes after placement (3 versus 10; 1.6 versus 5.5 percent). Conclusions are limited by the small number of events.
- In a 2017 randomized double-blinded trial comparing pre-procedural ultrasound versus landmark-based thoracic epidurals at T10 or higher, although the median number of needle passes was not significantly different between the two groups, postsurgical pain scores were significantly lower in the US group (3.0 ± 2.82 versus 4.7 ± 2.82 , respectively [$P = 0.015$]) suggesting that use of ultrasound to target appropriate intervertebral level may improve epidural quality.

Neuraxial procedure performance — The evidence regarding the technical benefits of using neuraxial ultrasound is mixed. Most, but not all, studies have found that preprocedural neuraxial ultrasound improves technical performance of neuraxial procedures in patients with difficult anatomy, but evidence of such benefits in patients with normal anatomy is less convincing [6-14]. In a 2023 network meta-analysis of 74 randomized trials (70 of the 74 studies involved lumbar neuraxial procedures) comparing landmark based neuraxial anesthesia versus preprocedural, real-time, or computer-aided ultrasound guidance, preprocedure and real time ultrasound improved first pass success in patients both with and without predicted anatomic difficulty [15]. Benefits of computer aided ultrasound were unclear.

- **Patients with difficult anatomy** – Several randomized trials have reported improved first attempt success and reduced needle reinsertion or redirection rates with ultrasound guidance for spinal or epidural anesthesia in patients with anticipated or known difficult anatomy; however, other studies have not found similar results [7-12,16]. Data regarding total procedure time are also inconsistent. As examples:
 - In a randomized trial of 120 orthopedic surgery patients who underwent spinal anesthesia and had predicted difficulty with the neuraxial procedure due to obesity, poorly palpable spinous processes, scoliosis, or prior spine surgery, preprocedure neuraxial ultrasound resulted in higher first attempt spinal procedure success (65 versus 32 percent) and fewer needle passes (median 6, interquartile range [IQR] 1 to 10 versus 13, IQR 5 to 21) compared with a landmark-based technique [8]. Total procedure time (ie, time to establish landmarks plus time to perform the spinal anesthetic) was longer when ultrasound was used (12.2 ± 6.0 versus 7.9 ± 7.7 minutes).
 - In a randomized trial of 80 parturients with obesity (body mass index [BMI] ≥ 30 kg/m²) who underwent spinal anesthesia for cesarean delivery, first attempt success rate was higher, and the number of needle passes and total procedure time were lower with preprocedure ultrasound compared with a landmark-based technique [7]. In a subgroup analysis of patients with BMI 30 to 35 kg/m², first attempt success, number of needle passes, and total procedure time were similar between groups.

- In contrast, in a randomized trial including 180 parturients with obesity who had spinal anesthesia for elective cesarean delivery, preprocedure ultrasound did not decrease the number of required needle passes, number of required skin punctures, time to complete the spinal procedure, or failed block [16].
- In a meta-analysis of 18 randomized trials including 1844 obstetric patients who underwent neuraxial anesthesia procedures, preprocedure ultrasound increased the first pass success rate in patients with predicted difficulty with the neuraxial procedure due to difficult landmark palpation, but not in patients with easily palpated anatomy [6]. Neuraxial ultrasound reduced the number of needle redirections (based on nine studies) and the number of skin punctures (based on seven studies).
- **Patients with normal anatomy** – Studies of the benefits of use neuraxial ultrasound in younger patients with normal spinal anatomy have been inconsistent.
 - A randomized trial of 150 parturients with easily palpable landmarks who underwent spinal anesthesia for cesarean delivery, average procedure time, number of skin punctures and needle passes, and successful blocks were similar in patients in whom preprocedure ultrasound was used versus a conventional landmark technique [17].
 - In two randomized trials of patients with easily palpable spinous processes who had labor epidural catheters placed for labor analgesia [18] or cesarean delivery [19], insertion time, first pass success, and number of attempts to thread the catheter were similar with preprocedure ultrasound versus conventional landmark palpation [18,19].
 - In a 2021 meta-analysis of 18 randomized trials including 1800 patients who underwent non-obstetric neuraxial anesthesia, preprocedure ultrasound increased the first attempt (skin puncture with or without needle redirection) success rate and decreased the need for three or more skin punctures and needle redirections, and reduced the incidence of bloody tap, compared with conventional landmark palpation [13]. First pass (single skin puncture without redirection) success rate was similar in the two groups, and neuraxial ultrasound increased the total time taken for the neuraxial procedure. In subgroup analysis, results were similar in patients with and without predicted difficulty with the neuraxial procedure.
- **Older adult patients** – Benefits of neuraxial ultrasound have been shown in older adult patients, who often present with challenging neuraxial procedures despite easily palpable spine landmarks. Anatomic features that may increase difficulty with neuraxial procedures in older adults include facet hypertrophy, interspinous/supraspinous calcification, and narrow interspinous spaces secondary to degenerative changes.
 - In a randomized trial of 100 older adult patients (mean age 65) who underwent spinal anesthesia for hip and knee replacements, preprocedure neuraxial ultrasound resulted in >50 percent reduction in number of passes for successful dural puncture compared with a landmark-based technique [11]. Out of the 100 patients enrolled, only three had a history of prior lumbar spine surgery and eight had palpable landmarks that were graded as difficult.
 - In another randomized trial of 80 older adults without known spine pathology or prior spine surgery who underwent spinal anesthesia for orthopedic surgery, use of preprocedure neuraxial ultrasound reduced the number of needle passes (median 1.0, IQR 1.0 to 2.0 versus 4.5, IQR 2.0 to 7.0), increased first pass success rate (65 versus 17 percent) and had reduced periprocedural discomfort scores (median 2, IQR 0 to 3 versus 5, IQR 2 to 6) compared with landmark-based techniques.

Accuracy in identifying intervertebral levels — Compared with landmark palpation, neuraxial ultrasound more accurately identifies a given spinal interspace. In the lumbar spine, the inter-cristal line (ie, the line between the iliac crests) is used as a rough guide for needle placement. This line crosses the L4 vertebral body or L4-5 interspace in many patients, though the line tends to be higher in females and in patients with obesity. (See "[Spinal anesthesia: Technique](#)", [section on 'Anatomy'](#).)

Multiple studies have found that landmark palpation is unreliable for determining the vertebral level, particularly in parturients, and clinicians often enter a higher interspace than intended during neuraxial anesthesia, thereby risking injury of the spinal cord. (See "[Serious neurologic complications of neuraxial anesthesia procedures in obstetric patients](#)", [section on 'Prevention of spinal cord and nerve root trauma'](#).)

In one study of 50 patients who underwent radiographs of the lumbar spine, both landmark palpation and neuraxial ultrasound were used to attempt to identify lumbar intervertebral spaces before the radiograph was performed [20]. Ultrasound imaging correctly identified the space in 71 percent of patients, whereas palpation was accurate in 30 percent. Importantly, all ultrasound estimates were within one level of the true interspace, whereas 27 percent of palpation assessments were inaccurate by more than one level higher or

lower. In another study designed to assess the learning curve for identification of the correct spinal level using neuraxial ultrasound (assessed by computed tomography), two clinicians achieved 90 percent accuracy after scanning 36 and 22 patients, respectively [21].

Anatomic anomalies of the spine including sacralization (fusion of L5 and S1 vertebrae) and lumbarization (S1 fails to fuse to rest of sacrum) can decrease the accuracy of determining the level in the lumbar spine by one vertebral level. The reported prevalence of such anomalies is as high as 35 percent, depending on the way it is diagnosed and the extent of the anomaly [22].

There are also challenges in precisely determining thoracic spine levels. This was demonstrated in a study of 210 patients in whom landmark identification of T7 was compared with chest radiograph confirmation [23]. The T7 spinous process was identified correctly 29 percent of the time using the C7 landmark (vertebra prominens) and 10 percent of the time using the inferior tip of the scapula landmark. There was a tendency to predict T7 to be lower than determined by radiograph, which could lead to common undesired outcomes, including inadequate analgesia and motor weakness.

Accuracy in measuring depth to the ligamentum flavum/dura complex — Several studies in pregnant and nonpregnant patients have found close agreement between ultrasound measured depth and actual needle insertion depth to the intrathecal or epidural space [24-27]. However, the depth measured using ultrasound should be considered an estimate. Depth tends to be underestimated because of tissue compression by downward force on the transducer.

In a 2016 meta-analysis of 13 studies of obstetric and nonobstetric surgical patients and one study in patients who underwent diagnostic lumbar puncture, the ultrasound measured depth to the epidural or intrathecal space was within 1 to 13 mm of the actual needle insertion depth [2]. In seven of the eight studies that reported a mean difference, the mean difference was ≤ 3 mm.

Measurements of depth in both the transverse and parasagittal oblique sagittal views have been comparable and can be used interchangeably as estimates of actual depth [28].

REAL-TIME NEURAXIAL ULTRASOUND

Successful use of real-time neuraxial ultrasound has been described in numerous studies [1,29-34]. Real-time neuraxial ultrasound is technically challenging, and study outcomes comparing its use with preprocedural ultrasound have been inconsistent. Further study is required before recommending routine use.

Most [15,35], though not all [33], randomized trials have found higher success rates with real time ultrasound, compared with landmark based blocks, though procedure times are longer with real time ultrasound. One randomized trial found that success rates for thoracic epidural placement were similar with real time ultrasound guidance compared with fluoroscopic guidance [36].

Results of studies comparing real-time ultrasound with preprocedural ultrasound are mixed.

- In a 2023 network meta-analysis of 74 randomized trials comparing landmark-based neuraxial anesthesia with preprocedural ultrasound guidance (57 trials), real-time ultrasound guidance (7 trials), or computer-aided ultrasound guidance, both preprocedural and real-time ultrasound improved first-pass success in patients, regardless of predicted anatomic difficulty [15]. The first-pass success rates were similar with preprocedural and real-time ultrasound, while the benefits of computer-aided ultrasound remained unclear.
- In contrast, a randomized trial involving 114 patients aged 70 years or older undergoing hip fracture surgery found that for spinal anesthesia, real-time ultrasound resulted in a lower first-pass success rate and a longer time from the start of the scan to obtaining cerebrospinal fluid compared to preprocedural ultrasound [34].

In a randomized trial comparing real-time ultrasound with fluoroscopic guidance for thoracic epidural placement, both techniques achieved similar first-pass success, final success, and postoperative opioid consumption [36]. However, real-time ultrasound guidance resulted in shorter times to identify the epidural space and complete the procedure.

SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See "[Society guideline links: Local and regional anesthesia](#)".)

SUMMARY AND RECOMMENDATIONS

- **Utility of neuraxial ultrasound** – Neuraxial ultrasound may be used to determine the intervertebral level for needle placement, identify the midline and optimal angle of needle insertion, and measure the depth to the epidural and intrathecal space. Ultrasound can also help assess anatomy that may make neuraxial procedures challenging (eg, spine rotation), and identify the most accessible interspace. (See '[Utility of neuraxial ultrasound](#)' above.)
- **Lumbar neuraxial ultrasound** – Lumbar neuraxial ultrasound is described briefly here, and in greater detail above. (See '[Ultrasound imaging](#)' above and '[Step-by-step lumbar ultrasound](#)' above.)
 - Position patients as they would be for the planned neuraxial procedure. (See '[Patient positioning](#)' above.)
 - Use a low frequency (2 to 5 MHz) curvilinear transducer, with the depth set to 7 to 10 cm. (See '[Equipment](#)' above.)
 - Obtain a parasagittal transverse process view of the lumbar spine ([figure 9](#)), followed by parasagittal articular process view ([figure 10](#)), and then the parasagittal oblique interlaminar view ([figure 11](#)). Clinicians experienced with neuraxial ultrasound may want to start with the parasagittal interlaminar oblique view. (See '[Sonoanatomy and transducer orientation](#)' above.)
 - Identify the sacrum and count sequential interspaces cranially ([figure 16](#)). Mark the desired interspace ([image 2](#)) and measure the depth to the posterior complex ([image 3](#)).
 - Rotate the transducer 90 degrees to obtain a transverse interspinous view ([figure 15](#)), visualizing the "bat sign" ([image 1](#)):
 - The posterior complex represents the bat's head.
 - The anterior complex represents the bat's bottom.
 - The transverse processes represent the wings of the bat.
 - Identify and mark the midline ([figure 17](#)). Measure the depth to the posterior complex ([image 4](#)).
 - Note the angle of the ultrasound beam that produces the clearest interlaminar view.
- **Thoracic neuraxial ultrasound** – Thoracic ultrasound below T9 is similar to lumbar ultrasound. The steps for a modified paramedian approach to mid to upper thoracic ultrasound (T4 to T9) are described briefly here, and in greater detail above. (See '[Step by step thoracic ultrasound](#)' above.)
 - Obtain a parasagittal view 5 cm lateral to the midline to identify the 12th rib, and count ribs while sliding the transducer cranially to the desired intervertebral level. Obtain a parasagittal transverse process view ([figure 19](#)), followed by a parasagittal oblique interlaminar view ([figure 12](#)).
 - Mark the intervertebral level with the widest interlaminar space and clearest view of the posterior and anterior complexes. Measure the depth from the skin to the posterior complex or lamina.
 - Rotate the transducer 90 degrees to obtain a transverse view ([figure 14](#)) and mark the midline.
 - Insert the epidural needle perpendicular to the skin approximately 0.5 cm lateral to and 1.5 cm inferior to the intersection of the marked horizontal and vertical lines to hit the lamina ([figure 20](#)). Walk the needle off the lamina caudally, with approximately 5 degrees of medial angulation, to reach the interlaminar space.
- **Ultrasound guidance versus landmark based neuraxial technique** – Preprocedure ultrasound may increase success and reduce the technical difficulty of neuraxial procedures, particularly for patients with difficult anatomy (eg, obesity, poorly palpable spinous processes, scoliosis, or prior spine surgery). Preprocedure ultrasound may be particularly beneficial in older adults who often have anatomic features that increase difficulty with neuraxial procedures despite having easily palpable spine landmarks. (See '[ultrasound guidance versus landmark based neuraxial techniques](#)' above.)

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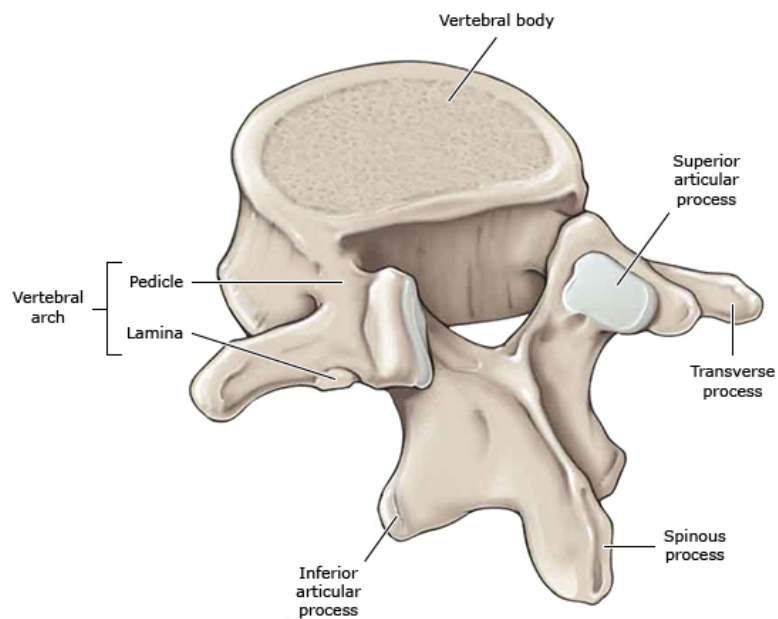
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Topic 106176 Version 11.0

GRAPHICS

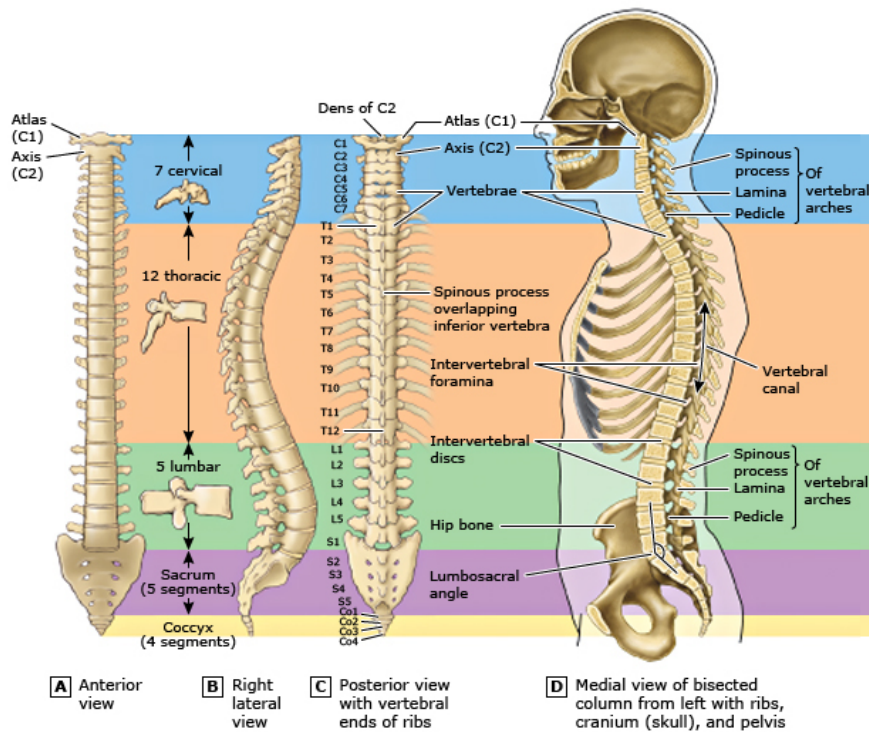
Figure 1: The vertebral body



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Figure 2: Spine anatomy overview



(A) This anterior view shows the isolated vertebral column.

(B) This right lateral view shows the isolated vertebral column. The isolated vertebrae are typical of each of the three mobile regions. Note the increase in size of the vertebrae as the column descends.

(C) This posterior view of the vertebral column includes the vertebral ends of ribs, representing the skeleton of the back.

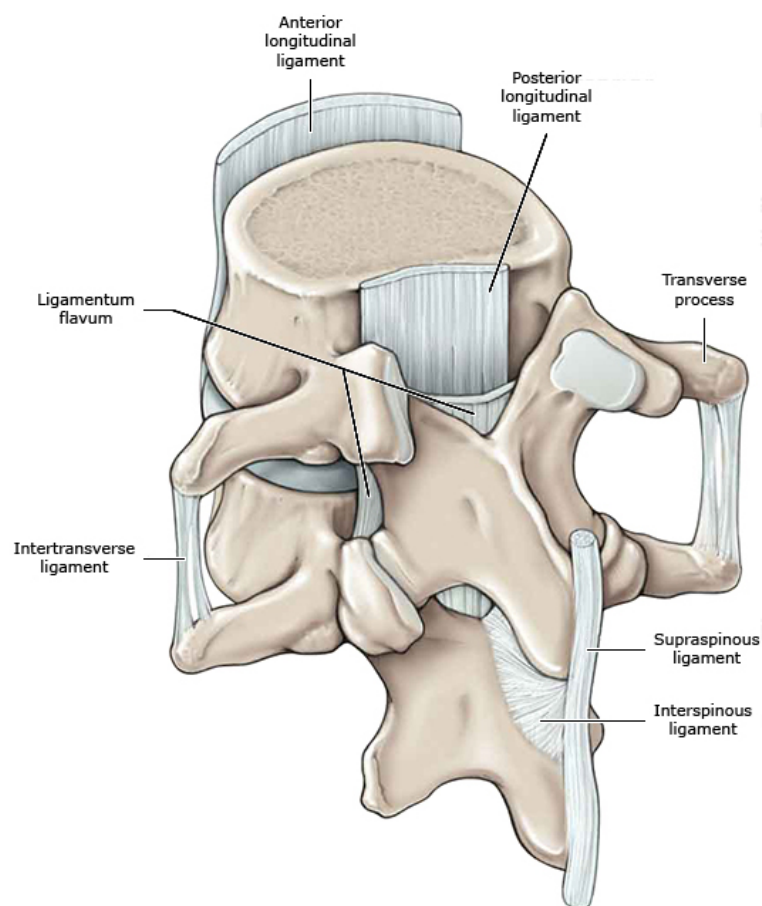
(D) This medial view of the axial skeleton in situ demonstrates its regional curvatures and its relationship to the cranium (skull), thoracic cage, and hip bone. The continuous, weight-bearing column of vertebral bodies and IV discs forms the anterior wall of the vertebral canal. The lateral and posterior walls of the canal are formed by the series of vertebral arches. The IV foramina (seen also in part B) are openings in the lateral wall through which spinal nerves exit the vertebral canal. The posterior wall is formed by overlapping laminae and spinous processes, like shingles on a roof.

IV: intervertebral.

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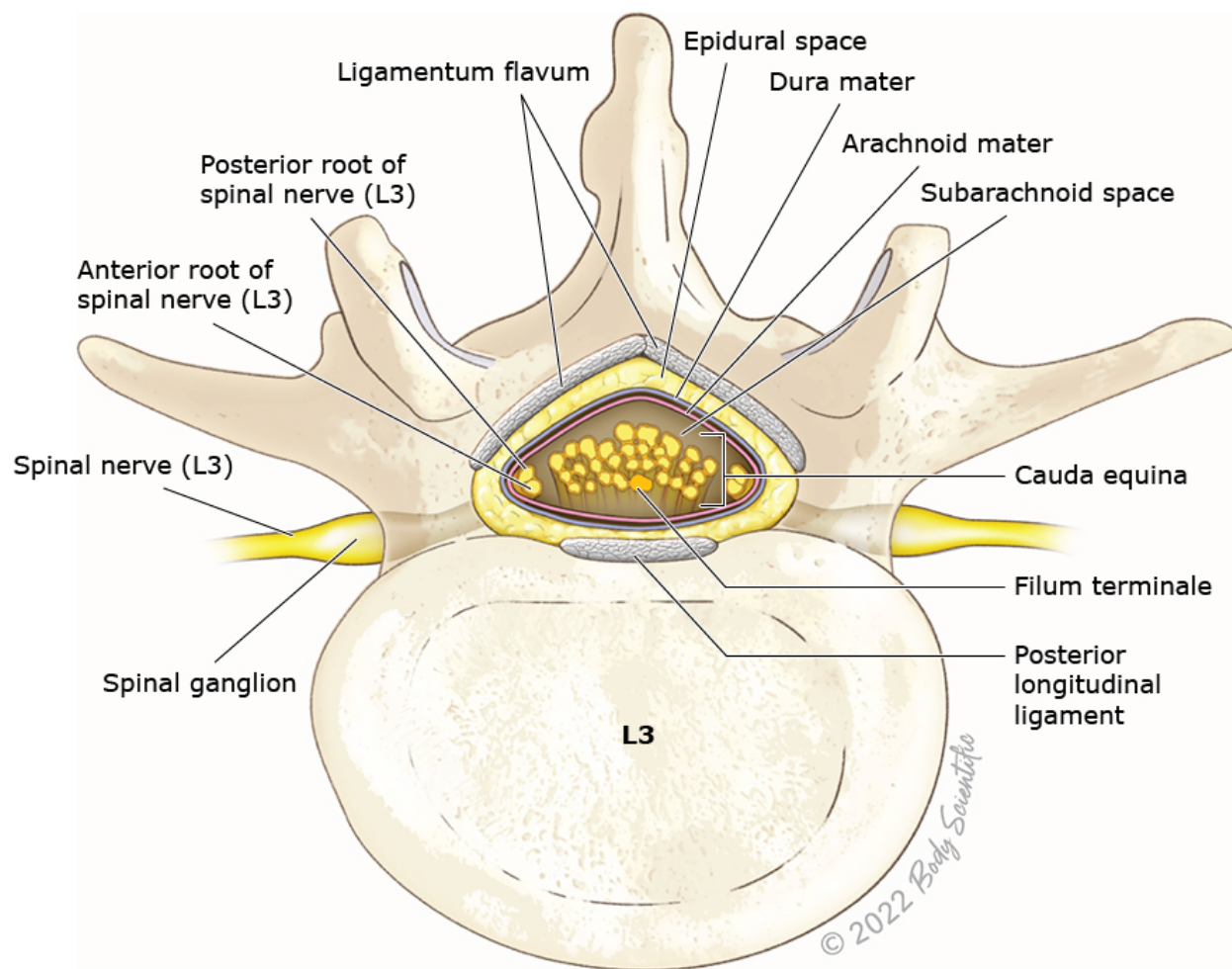
Figure 3: Vertebral bodies with ligaments



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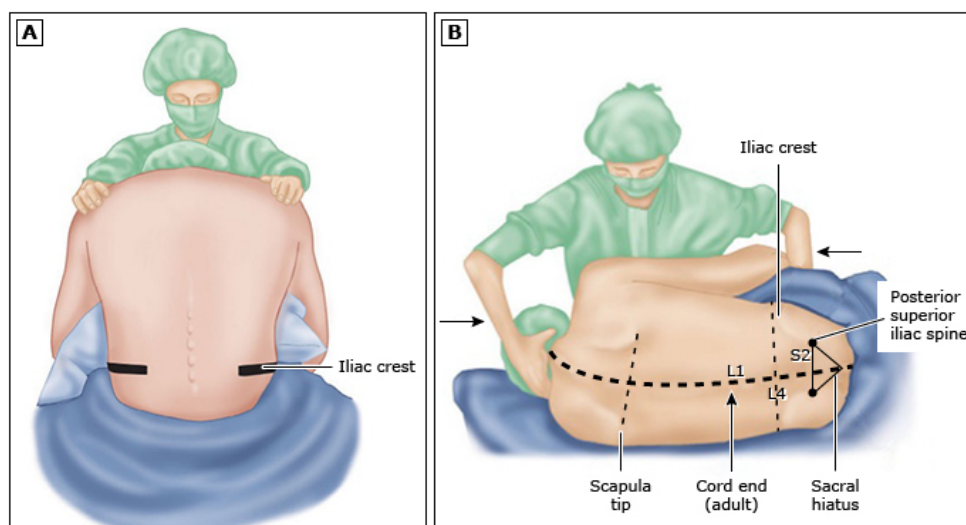
Figure 4: Spine cross section



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Figure 5: Positioning for neuraxial anesthesia



The sitting and lateral decubitus positions are used most commonly for spinal anesthesia. The sitting position may be particularly useful for larger patients, as the midline may be more easily estimated in this position if bony landmarks are not easily palpated. The patient should slouch symmetrically with shoulders over the hips to flex the spine.

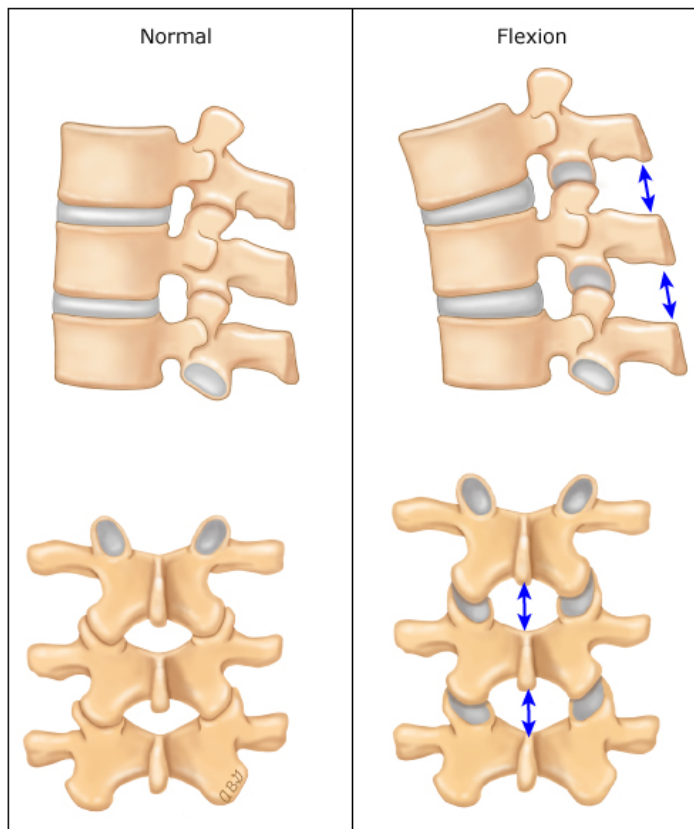
For the lateral decubitus position, the thighs should be drawn up with the hips maximally flexed and the lower back rounded or pushed out.

The intercrestal line (ie, a line between the iliac crests) is used as a rough guide for spinal needle placement. The needle should be inserted at or below this line to avoid spinal cord trauma.

For further details, refer to the topic on spinal anesthesia technique.

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Figure 6: Change in intervertebral space with spine flexion

For lumbar neuraxial anesthesia, the patient is optimally positioned with the lumbar spine flexed. As shown in this graphic, flexion opens the space for needle insertion between adjacent spinous processes. For further details, refer to the UpToDate topic on spinal anesthesia technique.

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Picture 1: Flexed seated positioning



If the procedural table is tilted downwards laterally to elevate the patient's knees relative to the buttocks, the patient will naturally lean forward and flex the spine. This opens the intervertebral spaces, thereby improving acoustic windows and the path for needle insertion.

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Figure 7: Epidural spinal positioning device

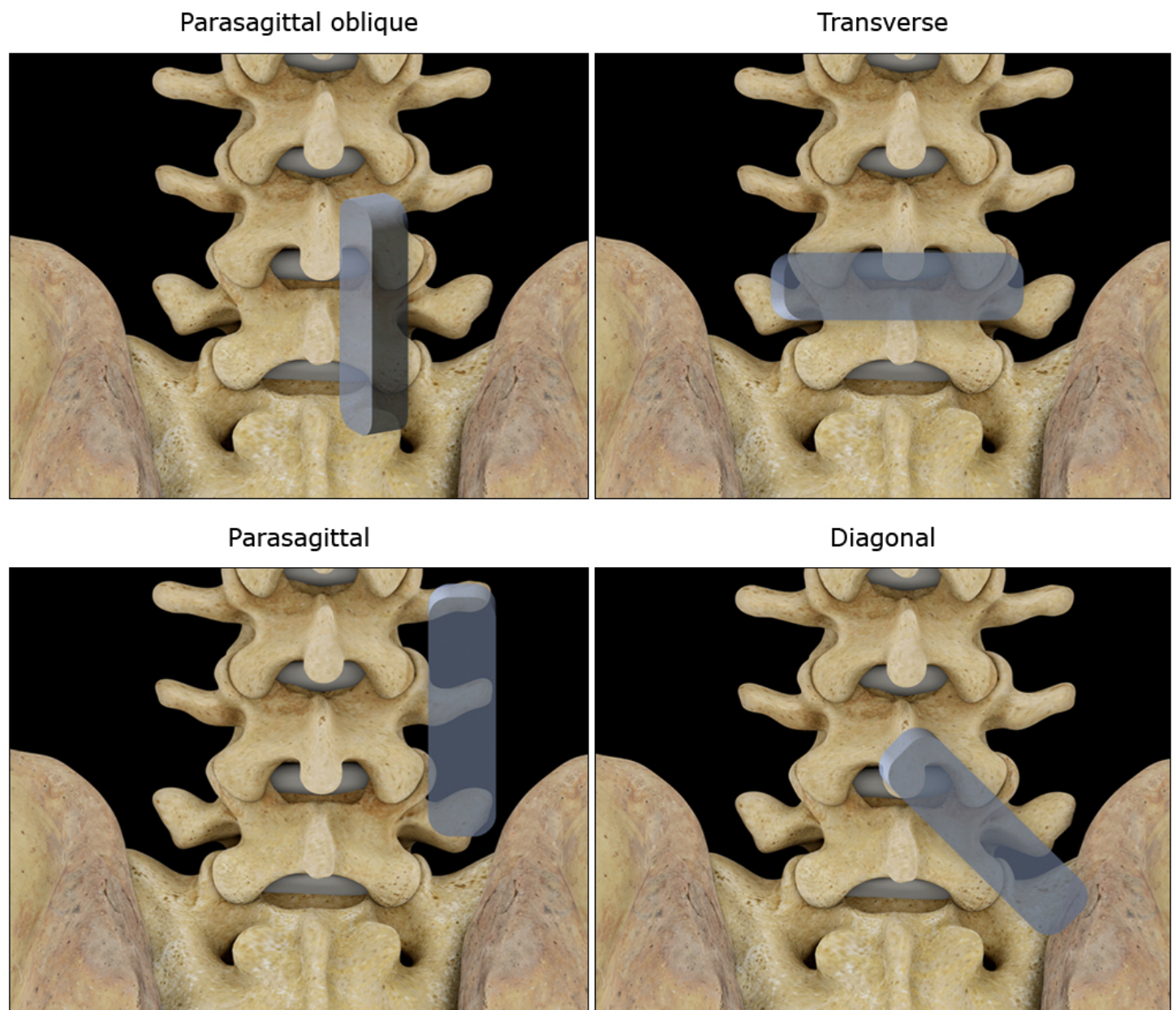


A neuraxial anesthesia positioning device such as the one shown can be used to provide stable patient support during spinal or epidural anesthesia placement.

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Figure 8: Transducer positions for neuraxial ultrasound

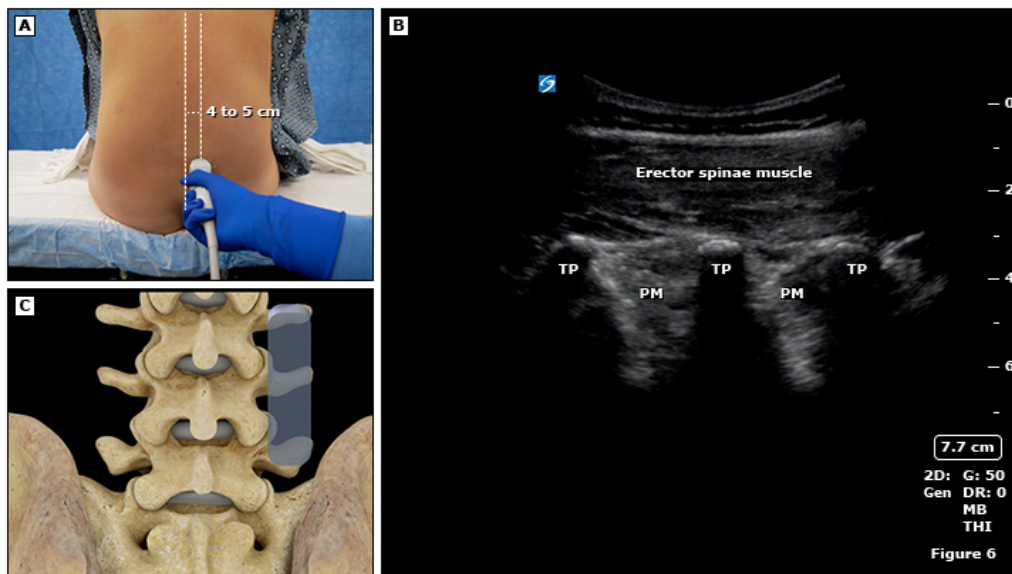


This graphic shows the various positions of the ultrasound transducer used for neuraxial ultrasound. For further detail refer to UpToDate content on neuraxial ultrasound for anesthesia.

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Figure 9: Transverse process view for lumbar neuraxial ultrasound



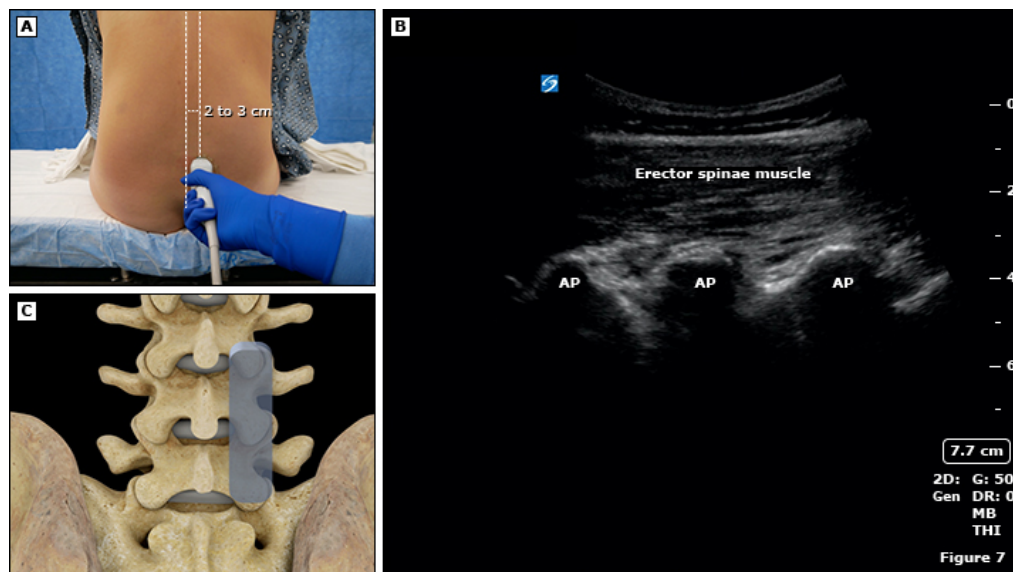
In the transverse process view, the transverse processes appear as hyperechoic, crescent-shaped reflections with hypoechoic, finger-like acoustic shadows below. The resulting pattern is referred to as the "trident," sign because of its similarity to the mythologic weapon. The psoas muscle is visualized between the transverse processes. For further detail refer to UpToDate content on neuraxial ultrasound for anesthesia.

TP: transverse process; PM: psoas muscle.

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Figure 10: Articular process view for lumbar neuraxial ultrasound



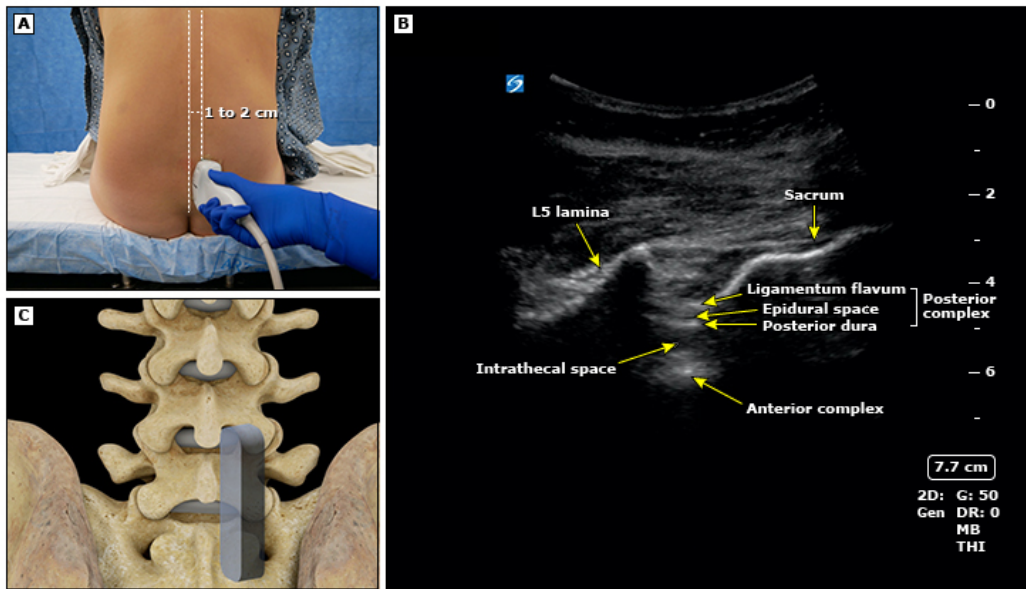
In the articular process view, a scalloped hyperechoic line is visualized, corresponding to the overlap of superior and inferior articular processes. This pattern is referred to as the "camel hump," sign. For further detail refer to UpToDate content on neuraxial ultrasound for anesthesia.

AP: articular process.

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Figure 11: Parasagittal oblique interlaminar view for lumbar neuraxial ultrasound

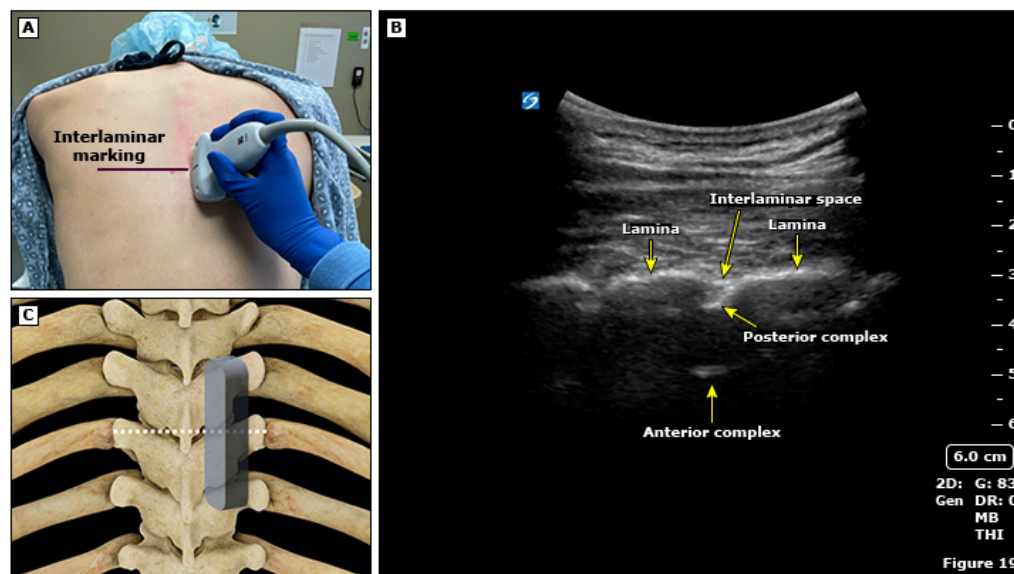


In the parasagittal oblique view, the interlaminar space is visualized and contains the posterior complex (ligamentum flavum/posterior epidural space/dura) and anterior complex (posterior longitudinal ligament/vertebral body). For further detail refer to UpToDate content on neuraxial ultrasound for anesthesia.

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Figure 12: Parasagittal oblique interlaminar view for thoracic neuraxial ultrasound

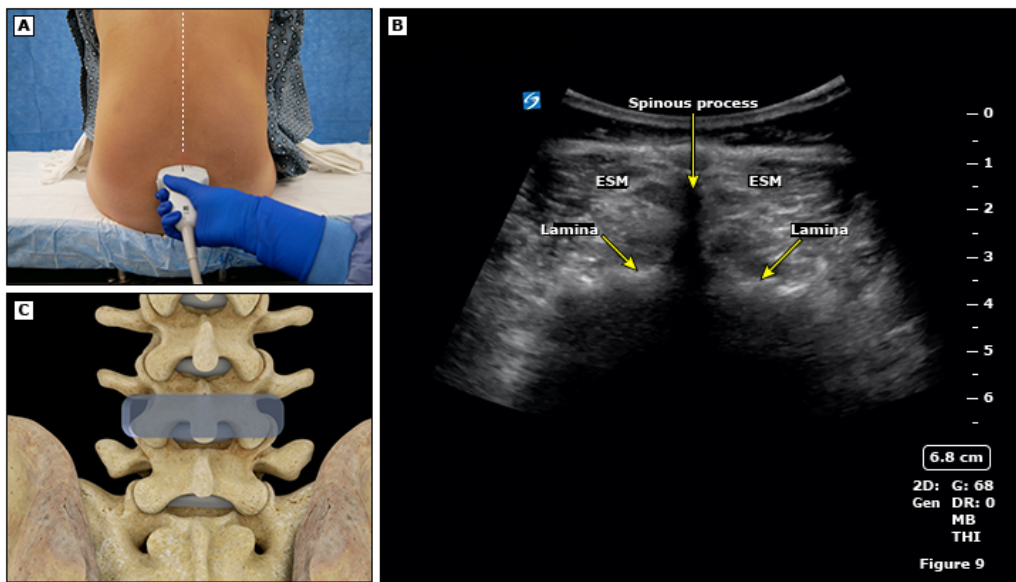


With the transducer centered on the interspace of interest, use the parasagittal oblique interlaminar view to identify the laminae, and if visible, the posterior and anterior complexes. This graphic is meant to be used in conjunction with UpToDate content on neuraxial ultrasound for anesthesia.

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Figure 13: Transverse scan, spinous process view for lumbar neuraxial ultrasound



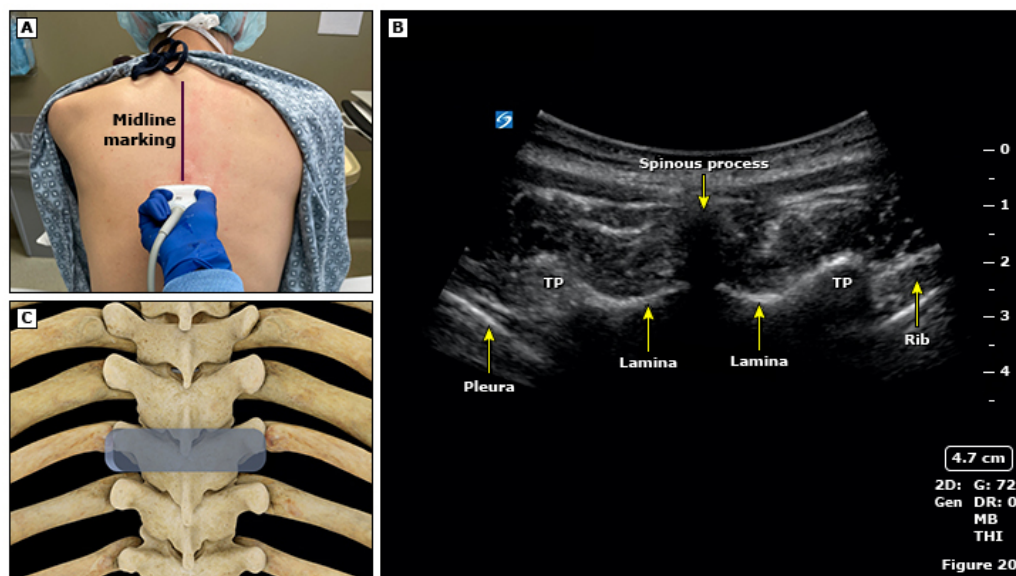
In this view, the tip of the spinous process and laminae appear hyperechoic, with large acoustic drop out below. For further detail refer to UpToDate content on neuraxial ultrasound for anesthesia.

ESM: erector spinae muscle.

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Graphic 138746 Version 1.0

Figure 14: Transverse scan, spinous process view for thoracic neuraxial ultrasound



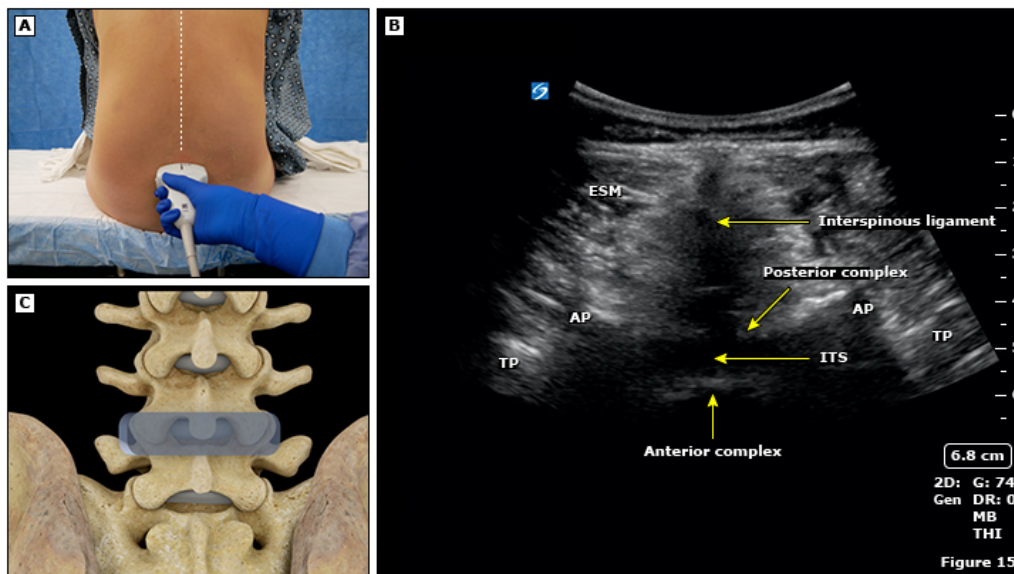
The transverse spinous process view can be used to identify the midline and to assess for spine rotation (eg, scoliosis). Spine rotation is suggested when lateral elements (eg, ribs, transverse processes, or laminae) appear at uneven depths from the skin.

TP: transverse process.

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Figure 15: Transverse scan, interspinous view for lumbar neuraxial ultrasound



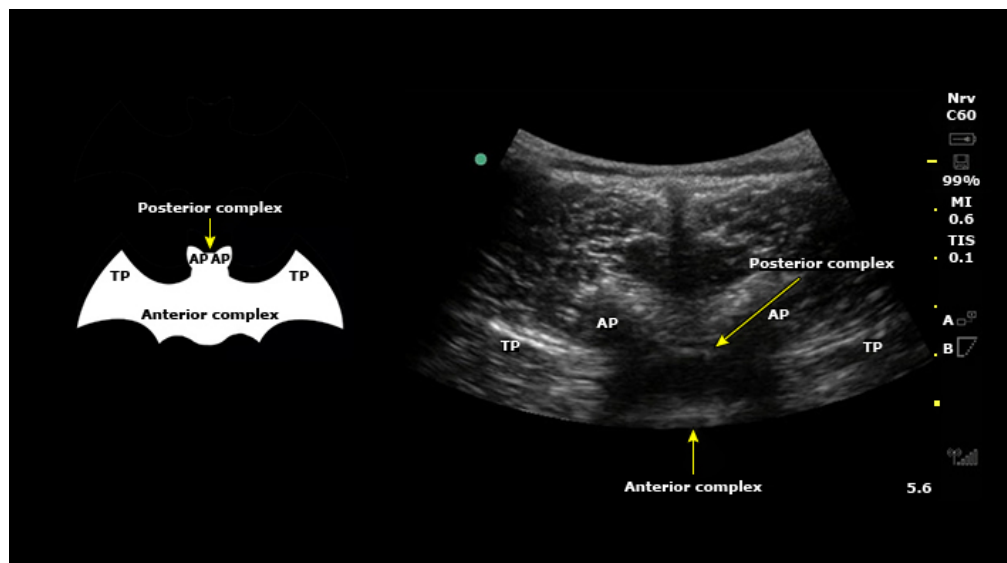
In this interspinous view, as is typical, visualization of the faint hyperechoic posterior complex band is difficult. The anterior complex band is thicker and easier to recognize. For further detail refer to UpToDate content on neuraxial ultrasound for anesthesia.

ESM: erector spinae muscle; AP: articular process; TP: transverse process; ITS: intrathecal space.

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Image 1: Bat sign in interspinous view for lumbar neuraxial ultrasound



The "bat," sign helps identify structures in the interspinous view for lumbar neuraxial ultrasound.

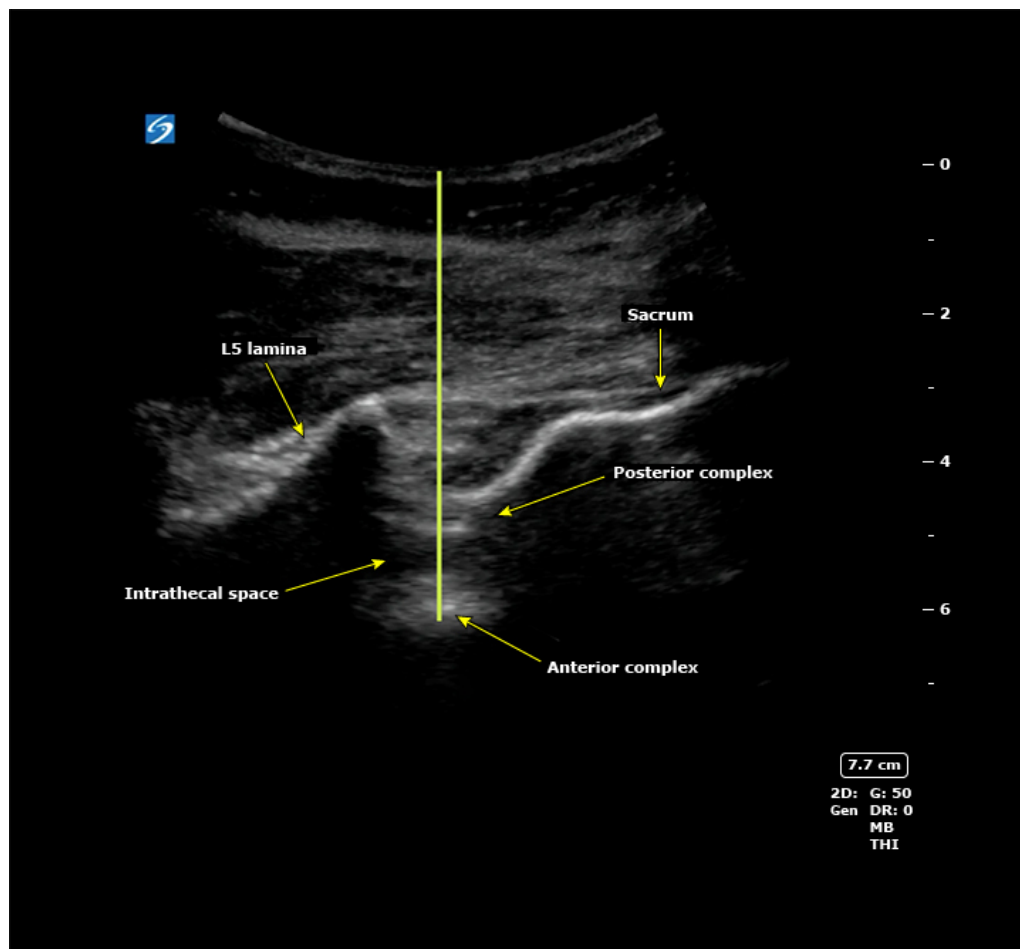
- The articular processes form the bat's ears.
- The posterior complex (ligamentum flavum, posterior epidural space, and posterior dura) forms the top of the bat's head.
- The transverse processes appear with an upward tilt that forms the bat's wings.
- The anterior complex (posterior longitudinal ligament/vertebral body) forms the bottom of the bat.

For further detail refer to UpToDate content on neuraxial ultrasound for anesthesia.

TP: transverse process; AP: articular process.

Graphic 138835 Version 1.0

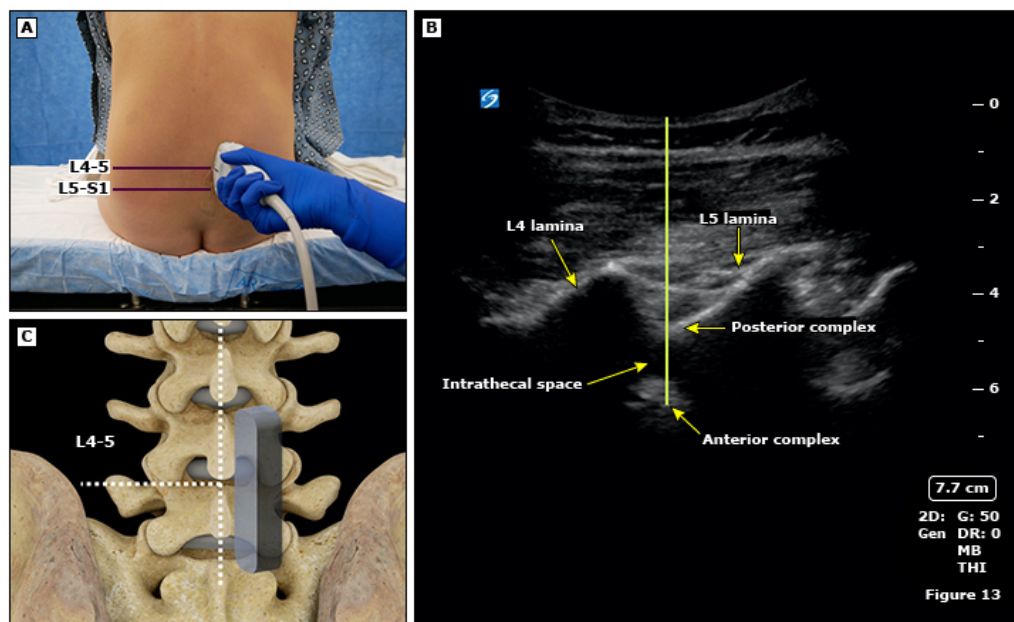
Image 2: Parasagittal oblique interlaminar view for lumbar neuraxial ultrasound with vertical center line



This image shows a vertical centered line that appears on many ultrasound screens with the M-mode or M/D cursor function. The vertical ultrasound line should be centered in the middle of the interlaminar space and corresponds to the midpoint of the long-edge of the ultrasound transducer. When counting interspaces, it can be useful to mark the skin at the center of each interspace. For further detail refer to UpToDate content on neuraxial ultrasound for anesthesia.

Graphic 138836 Version 1.0

Figure 16: Counting interspaces during lumbar neuraxial ultrasound

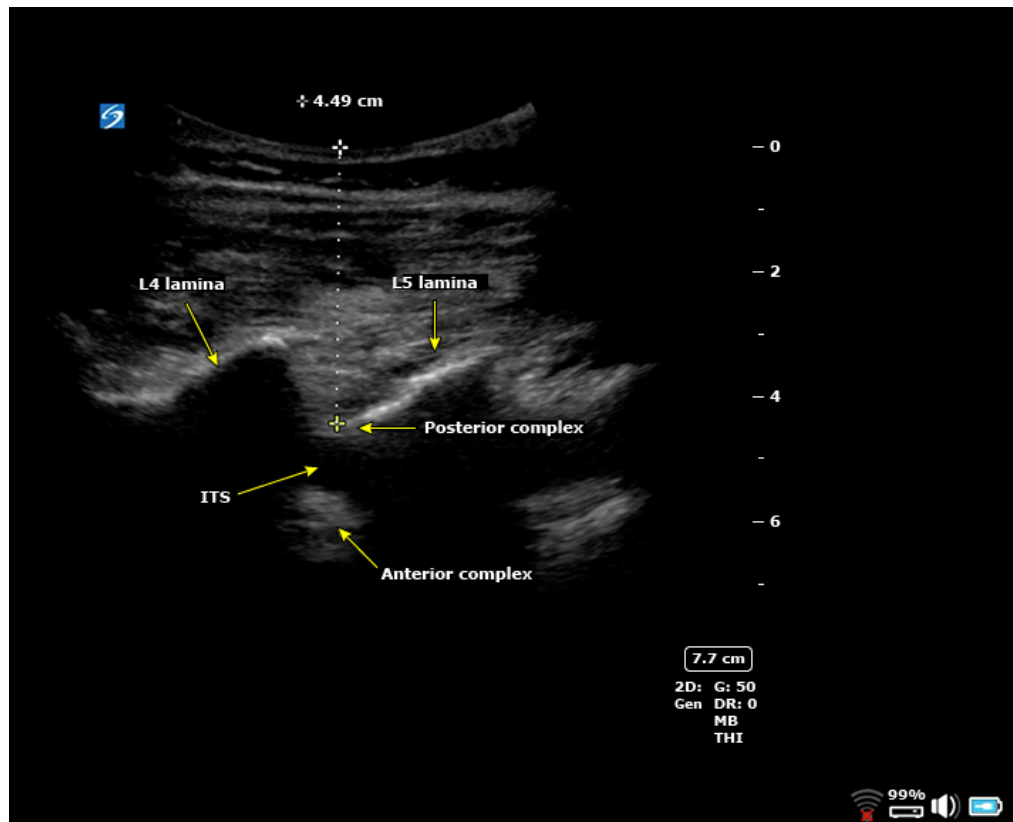


When counting interspaces to determine the target intervertebral level for neuraxial anesthesia, start by identifying the sacrum and the L5-S1 interspace. The vertical ultrasound line corresponds to the midpoint of the long edge of the ultrasound transducer and appears on many ultrasound machines when the M-mode or M/D cursor function is turned on. Slide the transducer cephalad centering the vertical ultrasound line in the midpoint of each interlaminar space, making horizontal skin markings until the desired interspace is reached. For further detail refer to UpToDate content on neuraxial ultrasound for anesthesia.

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Image 3: Measuring the depth to the posterior complex during lumbar neuraxial ultrasound, parasagittal oblique interlaminar view

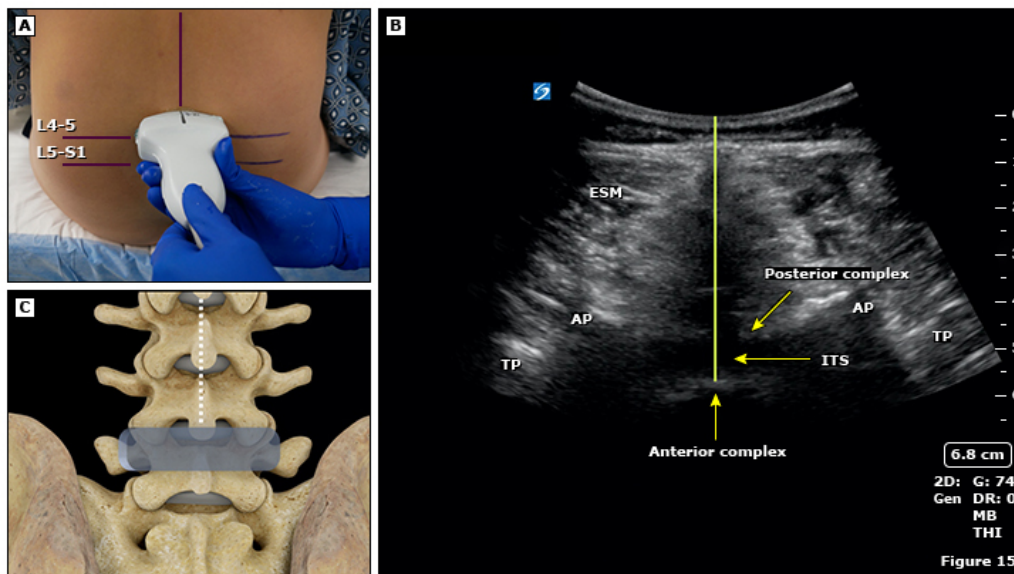


In the parasagittal oblique interlaminar view, use the caliper function on the ultrasound machine to measure the depth from the skin to the posterior complex (ligamentum flavum, epidural space, and dura). For further detail refer to UpToDate content on neuraxial ultrasound for anesthesia.

ITS: intrathecal space.

Graphic 138837 Version 2.0

Figure 17: Identifying the midline during lumbar neuraxial ultrasound, transverse interspinous view



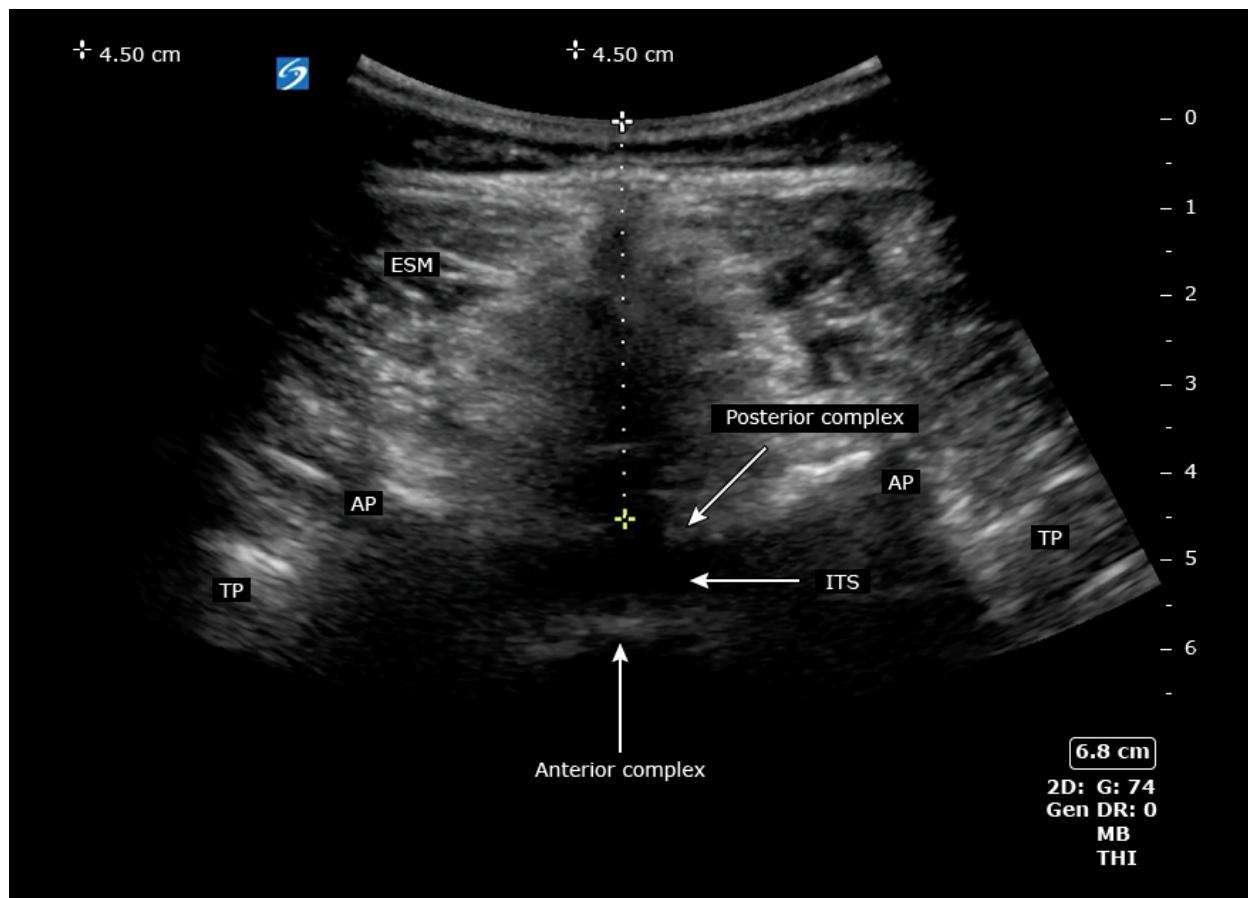
Center the ultrasound image. The line corresponds to the midpoint of the long edge of the transducer and appears when the M mode or M/D cursor function is turned on. For further detail refer to UpToDate content on neuraxial ultrasound for anesthesia.

ESM: erector spinae muscle; AP: articular process; TP: transverse process; ITS: intrathecal space.

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Image 4: Measuring the depth to the posterior complex during lumbar neuraxial ultrasound, interspinous view

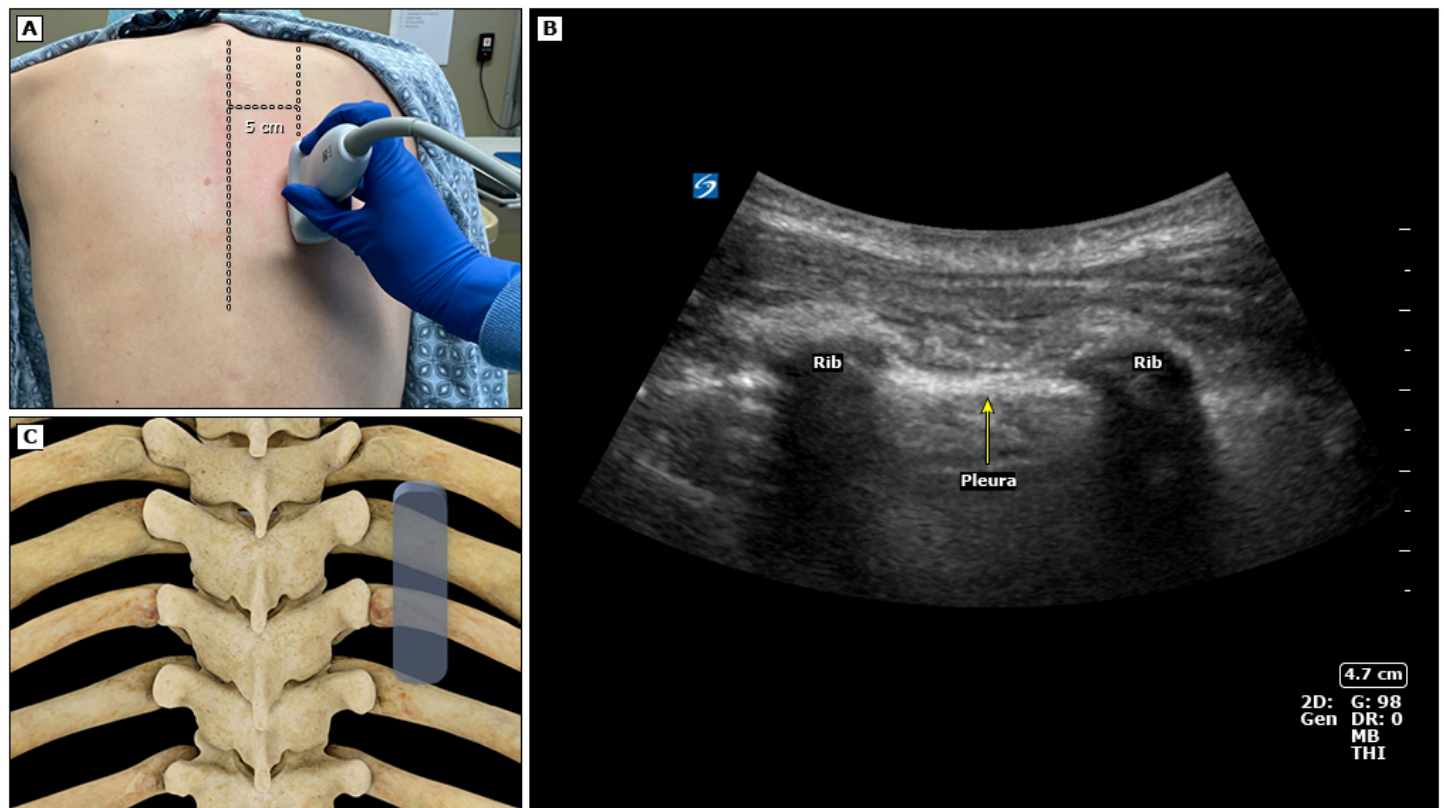


In the interspinous view, use the caliper function on the ultrasound machine to measure the depth from the skin to the posterior complex (ligamentum flavum, epidural space, and dura). For further detail refer to UpToDate content on neuraxial ultrasound for anesthesia.

AP: articular process; ITS: intrathecal space; TP: transverse process; ESM: erector spinae muscle.

Graphic 139699 Version 1.0

Figure 18: Parasagittal view of ribs for thoracic neuraxial ultrasound

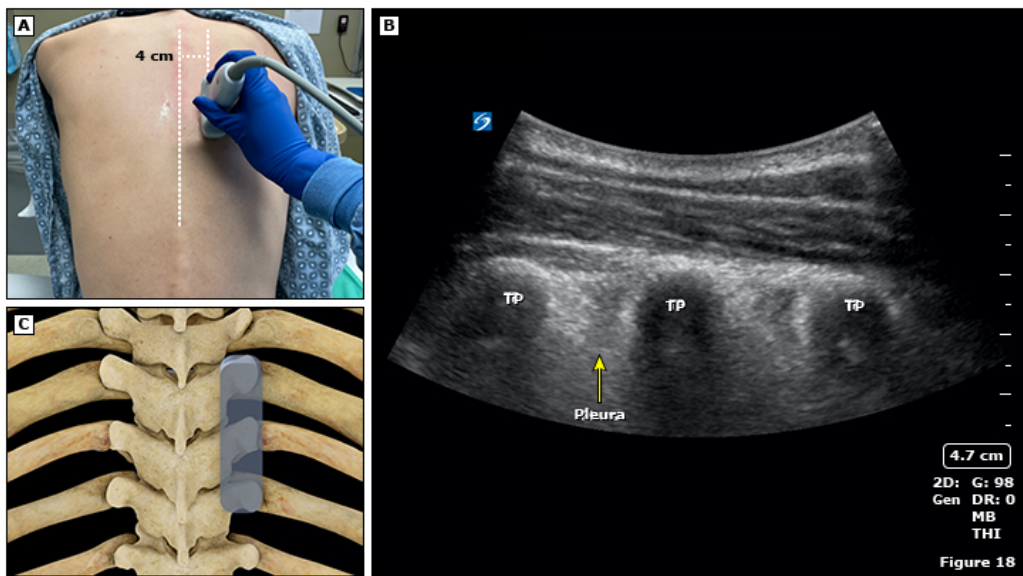


Position the ultrasound transducer in a parasagittal orientation 5 cm lateral to the midline to identify the 12th rib, then count cephalad to the desired interspace. The shimmering hyperechoic pleura is seen between the ribs. For further detail refer to UpToDate content on neuraxial ultrasound for anesthesia.

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Figure 19: Parasagittal transverse process view for thoracic neuraxial ultrasound



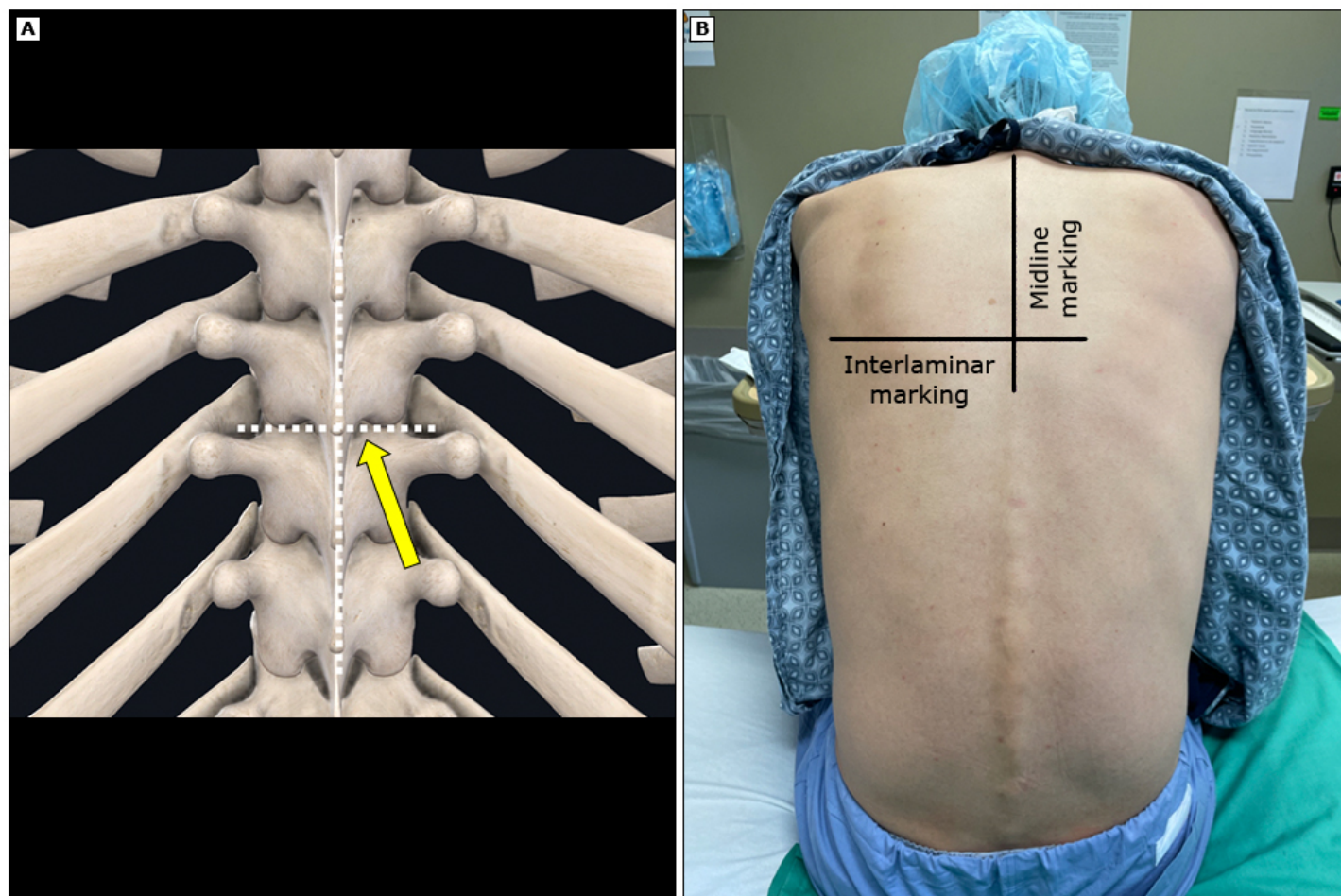
This image shows the transverse processes, and the pleura, which is sometimes difficult to visualize in this view. For further detail refer to UpToDate content on neuraxial ultrasound for anesthesia.

TP: transverse process.

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Figure 20: Paramedian approach to thoracic epidural after neuraxial ultrasound



This arrow in the graphic on the left shows the needle insertion site and angle for paramedian thoracic epidural needle placement after preprocedure neuraxial ultrasound. The midline and interlaminar space for the desired interspace are determined using ultrasound, and the skin is marked with corresponding vertical and horizontal lines. The needle should be inserted approximately 0.5 cm lateral to and 1.5 cm inferior (caudal) to the intersection of the two lines, aimed medially and at an angle of approximately 5 degrees from perpendicular to the skin. This graphic should be used in conjunction with UpToDate content on neuraxial ultrasound for anesthesia.

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Graphic 138762 Version 1.0

Contributor Disclosures

Brinda B Kamdar, MD Consultant/Advisory Boards: Rivanna Medical [Medical imaging technology]. All of the relevant financial relationships listed have been mitigated. **David Hao, MD** No relevant financial relationship(s) with ineligible companies to disclose. **Robert Maniker, MD** No relevant financial relationship(s) with ineligible companies to disclose. **Nancy A Nussmeier, MD, FAHA** No relevant financial relationship(s) with ineligible companies to disclose.

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